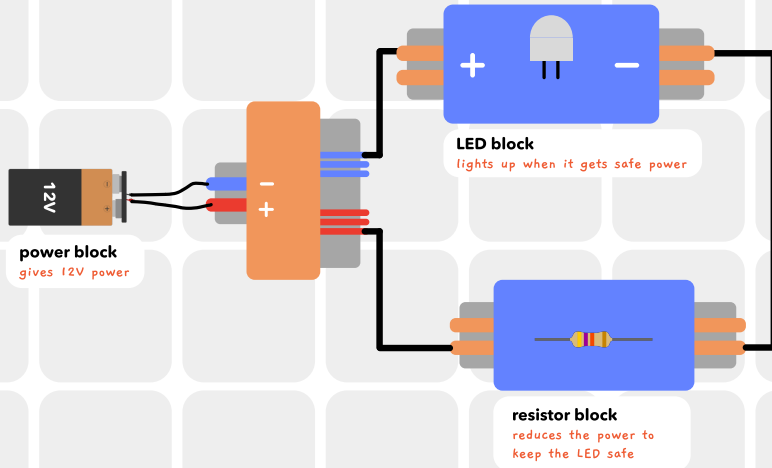


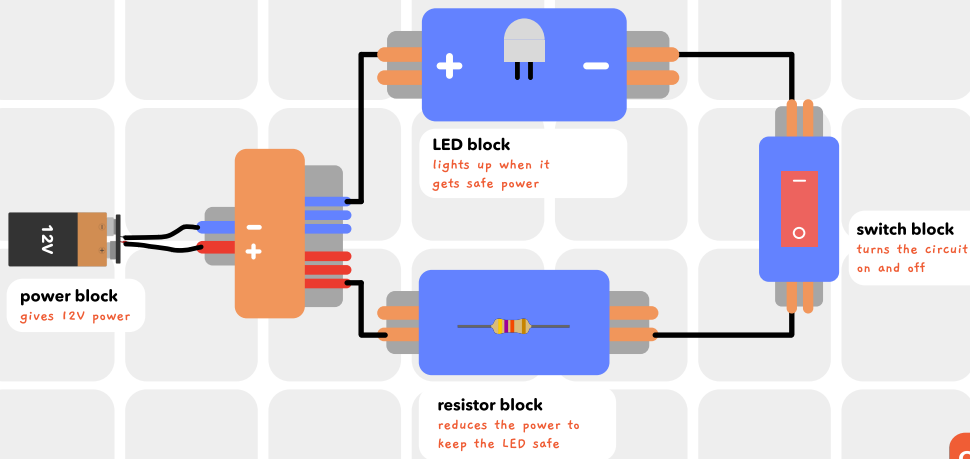
circuit 1 : resistor block & led block

Task: Connect all blocks as per the circuit below such that the LED lights up when it gets safe power.



circuit 2 : switch block

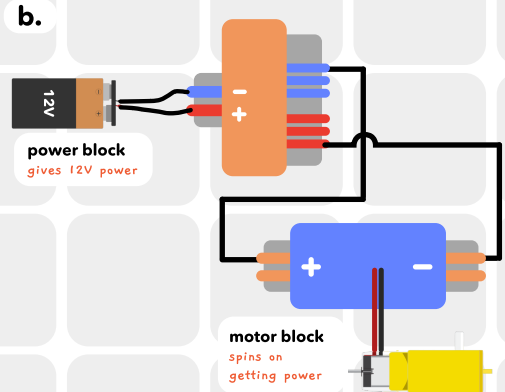
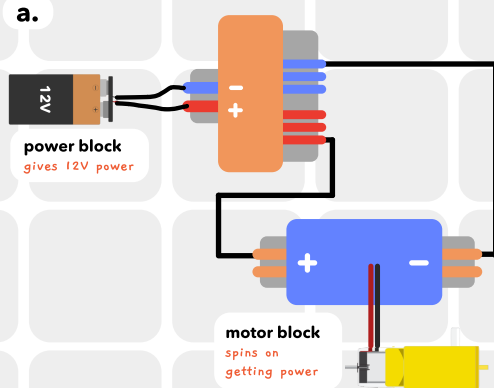
Task: Connect all blocks in a circuit and place the switch such that the LED turns on when the switch is ON.



circuit 3 : motor block

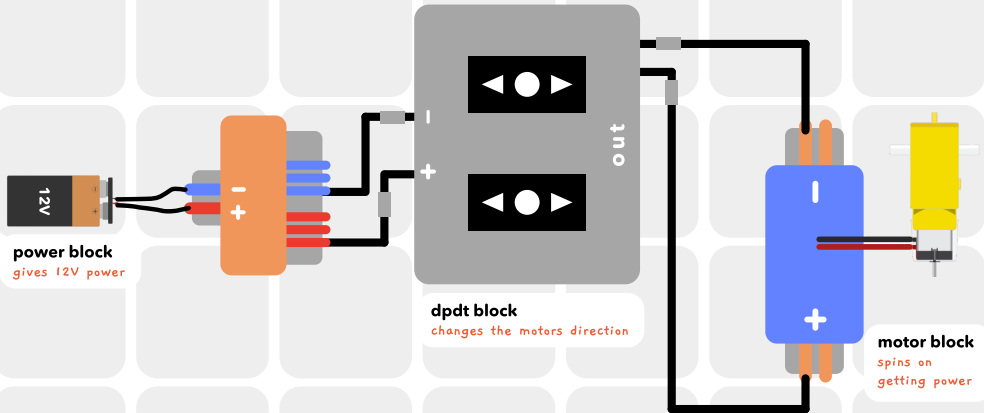
Task A: Connect the battery to the motor and observe the motor spin.

Task B: Reverse the wire connections and observe how the motor spins in the opposite direction.



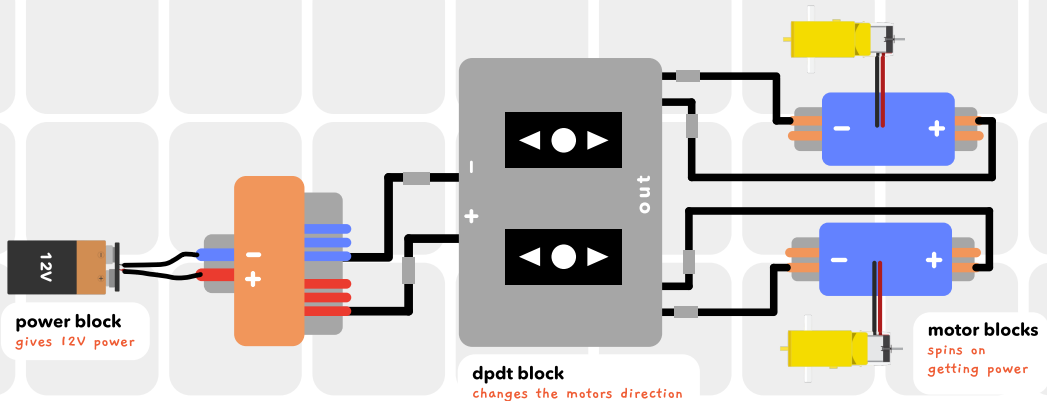
circuit 4 : dpdt block

Task: Connect the battery and motor through the DPDT block.
Press the DPDT switch to change the direction of the motor.



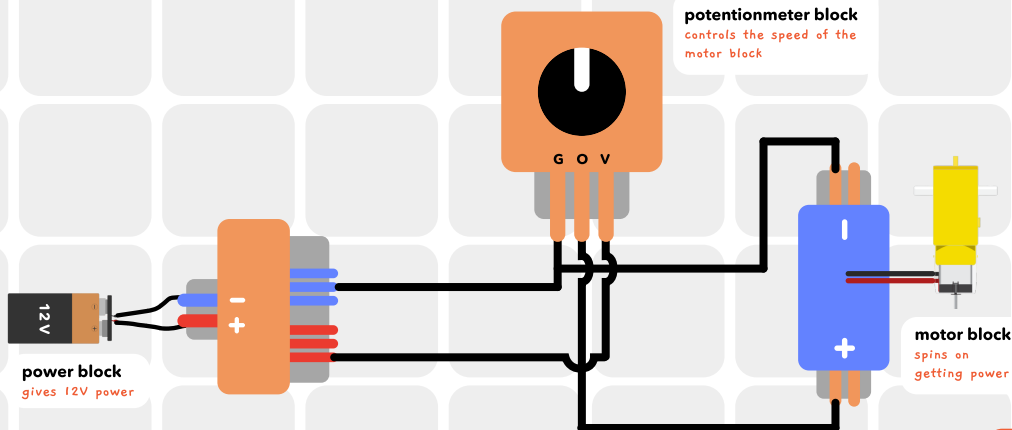
circuit 5 : dual dpdt circuit

Task: Connect each motor to its own DPDT switch and to the battery.
Press each DPDT switch to control each motor's direction.



circuit 6 : potentiometer block

Task: Connect the battery and motor through the potentiometer block.
Turn the knob to control the speed of the motor.



circuit 7 : motor driver + ir sensor

Task: Connect the IR sensor to the motor driver and connect the motor to the driver outputs. The IR sensor's signal will tell the motor driver when to run the motor, such that the motor moves only when the IR sensor is activated.

